

Building Energy systems Course Project



Please write your names and birth year in your reports. The assignments need to be submitted individually. **Deadline for delivering the results is 17 Nov 2026.**

The input data is based on your date of birth: , , , , .

Example: 1985, 10, 29: a=1, b=0, c=2, d=9, ab=10, cd=29.

Please assume the following:

- a) A housing association in the X-city (Choose the city number from the attached list) has problems with the heat supply in one of their multi-family houses. When it gets really cold, indoor temperatures drop below 20 °C. As a HVAC consultant, you are instructed to calculate the power that the radiator system must supply to the building in design conditions, where the aim is to have an average indoor temperature of $(20 + \frac{c}{10})$ °C. Calculate the balance temperature (T_b). What will be the **needed power supply** without considering the internal heat generation? Use the DVUT method, you can find the related DVUT temperature from the attached excel file “Ind-1-Värmeeffektbehov-MB3-beräkningsverktyg-191001” from SGBC Miljöbyggnad indicator 1.

Hint: in the power calculation the aim is to design a heating systems based on the worst case scenario, i.e. without counting in the solar radiation. Here for simplicity you can even ignore the internal heat from the people, and only assume the heat contribution from the appliances $(2 + \frac{c}{10})$ W/m² Atemp.

- b) Calculate the **average heat transfer coefficient**, i.e. U_m .

- c) Draw the **duration diagram** for the X-city (where building is located) and calculate the heat deficit between mid-September until mid-May. You can assume the solar irradiation, I , from Figure 1 for all cities. Use the weather file for the year (2015 + $\boxed{a+a+c}$) from one of the following links:

2015: <http://www.sveby.org/wp-content/uploads/2016/06/Klimatdatafiler-f%C3%B6r-2015.zip>
 2016: <http://www.sveby.org/wp-content/uploads/2018/06/Klimatdatafiler-f%C3%B6r-2016-v2.0.zip>
 2017: <http://www.sveby.org/wp-content/uploads/2018/06/Klimatdatafiler-f%C3%B6r-2017-v2.0.zip>
 2018: <http://www.sveby.org/wp-content/uploads/2019/01/Klimatdatafiler-f%C3%B6r-2018-v1.0.zip>
 2019: <http://www.sveby.org/wp-content/uploads/2020/01/Klimatdatafiler-f%C3%B6r-2019-v1.0.zip>
 2020: <http://www.sveby.org/wp-content/uploads/2021/01/Klimatdatafiler-for-2020-v1.0.zip>

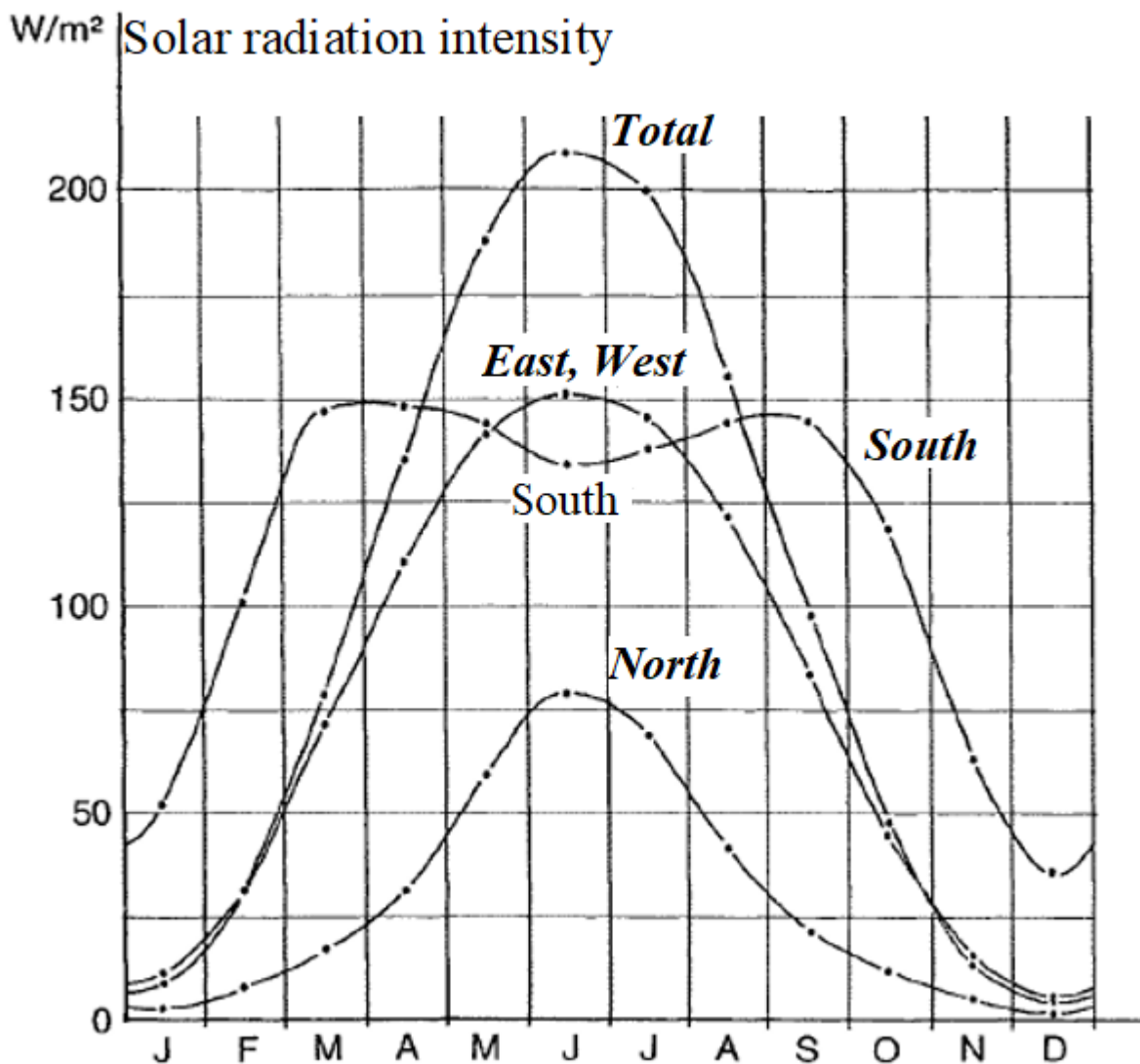


Figure 1: Solar radiation intensity for Örebro (60° latitude) for different directions, monthly averaged throughout the year [reference: Nevander L.E., Elmarsson B. (1994): *Fukthandboken – Praktik och teori*]. Hint: for energy calculations, you can multiply the data from the diagram by the number of hours during each month.

- d) Calculate the primary energy number, correct your results (energy use) based on the typical year climate data, (see Normal year correction in the course lectures), and find the energy class of the whole building, based on the heating system type $\boxed{c}$. You can

find the typical year outdoor temperature from the links below:

https://www.sveby.org/typar_1991_2020_smhi_bov/

Please motivate your results and suggest improvement measures to reduce the heating demand.

The housing association's board has compiled certain documentation in the form of drawings, measurement protocols and design data; if other parameters are needed please assume your own numbers and motivate your choice. You will get the following information:

- The building is $(3 + \boxed{a})$ stories high and has a floor plan of $10 \times 45 \text{ m}^2$. Each story has a height of 2.5 m.
- The foundation construction including the ground has a calculated U value corresponding to $0.23 \text{ W}/(\text{m}^2 \cdot \text{K})$. The concrete slab is 200 mm thick.
- The intermediate floors are made of concrete, each have a thickness of 160 mm.
- According to the OVK (Mandatory Ventilation Control, which all multi-family houses regularly go through) the ventilation flow rate is sufficient to keep the CO_2 level below 1000 ppm. Assume $(30 + \boxed{ab})$ people, inside the whole building, each producing 0.005 l/s of CO_2 and the CO_2 concentration in outdoor air to be 350 ppm. The ventilation system is mechanical exhaust type.
- The roof consists of 120 mm concrete with overlaid thermal insulation and suspended outer roof. It's U-value is $0.16 \text{ W}/(\text{m}^2 \cdot \text{K})$.
- The total window area is $(100 + \boxed{cd}) \text{ m}^2$ and consists of 2-glazed windows wooden frames (U-value $2.9 \text{ W}/(\text{m}^2 \cdot \text{K})$ may be assumed). Each window has the solar heat gain coefficient, g, of 0.7, and solar radiation factor, K_{rad} , of 0.5. 10 % of the total windows are located towards south and north and 40% towards east and west.
- An investigation indicates that on each floor there are approximately 130 meters of internal load bearing walls composed of concrete, 160 mm thick, and about 120 meters non-supporting walls made up of wood stud walls with 13 mm plasterboard as cladding. The thermal storage of wood stud can be neglected. All walls contribute to the thermal mass of the building.
- The external walls consist mainly of $\frac{1}{2}$ -bricks width (12 cm), 25 mm ventilated air gap, asbestos board, 95 mm mineral wool between wood studs (95 x 45 mm and center to center distance of 600 mm), PE-foil (moisture retarder) and 13 mm gypsum board. The thermal resistance for the ventilated air gap with the connected outdoor layer can be assumed as $0.2 \text{ m}^2 \cdot \text{K}/\text{W}$. The thermal impact of asbestos board and PE-foil is neglected. The U-value of the wall is estimated using the λ -value method, see the appendix A
- Along each long side of external wall there are exterior corridors (walking balconies/extended intermediate floors), which are serious thermal bridges with a linear heat loss coefficient, Psi-value, of $0.6 \text{ W}/(\text{m} \cdot \text{K})$.
- There are $(15 + \boxed{d})$ doors with U-value $2.0 \text{ W}/(\text{m}^2 \cdot \text{K})$. Each door area is 2.1 m^2 .

- The air leakage is estimated to be 0.05 ACH (1/h).
- Internal heat generation from household appliances is assumed to be $(2 + \boxed{C})$ W/m². Impact of human presence and solar irradiation is neglected in the power supply assessment and in the calculation of the balance temperature, (T_b).
- Assume that the density and specific heat capacity of gypsum (plaster) is 650 kg/m³ and 720 J/(kg*K) respectively.
- For concrete the corresponding values are 2300 kg/m³ and 900 J/(kg*K), respectively.
- The heating system type $\boxed{C}$ is (0=district heating, 1=direct electric element, 2=air to air heat pump with COP=3 and 3=air to air heat pump with COP=5).

Please deliver your report (send email to Sana.Sayadi@hig.se) in a typed format, unless if you deliver by hand writing it should be very readable. The deadline is **17 Nov 2026**. You should write and discuss about your results as well as specifically mention the following outputs:

Your input:

Your city
The DVUT
Tin

Your results:

Average heat transfer coefficient, U_m :
Balance temperature, T_b :
Required heating system power:

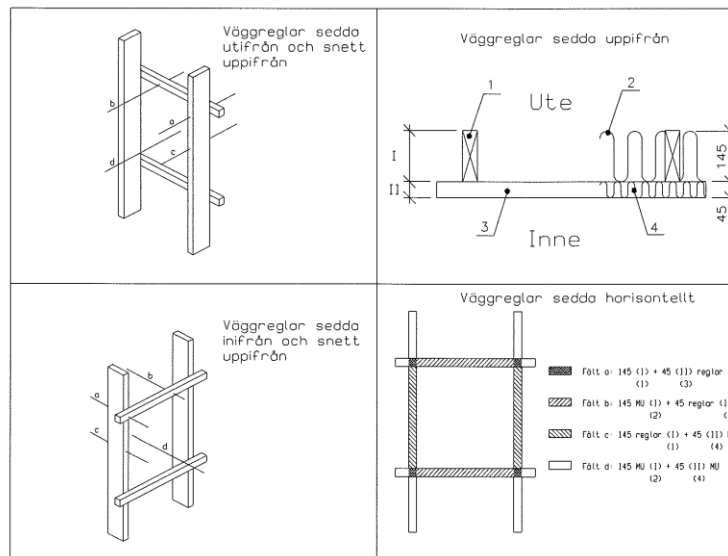
Heat gain via people
Heat gain appliances
Solar Heat gain
Heat losses via transmission
Heat losses via ventilation
Heat deficit

Duration diagram
Heating season Degree hours (mid-September until mid-May)
The primary energy number and the Energy class of the whole building (after **Normal year correction**)

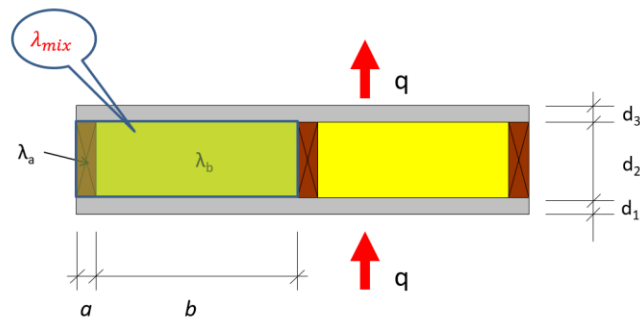
Explain the steps and discuss your results and describe in your own words, what you understand from the results?

Appendix A:

The insulation within the building wall is set on wooden frame, similar to the figure below:



Thus, the layer consists of both insulation and wooden beams. The thermal conduction of the mixture layer and the enclosing layers can be calculated by the so-called λ -value method according to the equation below:



$$\lambda_{mix} = \frac{a}{a+b} \cdot \lambda_a + \frac{b}{a+b} \cdot \lambda_b$$

$$R_{tot} = R_{si} + \frac{d_1}{\lambda_1} + \frac{d_2}{\lambda_{mix}} + \frac{d_3}{\lambda_3} + R_{se} \quad U_\lambda = \frac{1}{R_{tot}}$$